

Smartvision System Manual

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1 Introduction

2 Configuration

Smartvision System behavior can be configured from the *config.json* file inside the main folder. The configuration file has the following parameters:

```
{
  "width": 1280,
  "height": 720,
  "fps": 30,
  "max_zoom_factor": 5.0,
  "model_path": "./models/yolov8n.rknn",
  "i2c_device": "/dev/i2c-8",
  "i2c_addr": 64,
  "pan_max": 170,
  "pan_min": 10,
  "tilt_max": 170,
  "tilt_min": 10,
  "pan_base": 90,
  "tilt_base": 90,
  "pan_kp": 21.0,
  "pan_ki": 0.0,
  "pan_kd": 0.5,
  "tilt_kp": 14.0,
  "tilt_ki": 0.0,
  "tilt_kd": 0.3,
  "gimbal_deadzone": 0.03,
  "auto_return_to": 3000,
  "osd_enabled": true,
  "osd_color": [
    0,
    255,
    0
  ],
  "osd_thickness": 1,
  "osd_scale": 1.0,
  "osd_pos_x": 10,
  "osd_pos_y": 30
}
```

A description of each parameter can be found in the table below.

Field	Type	Min	Max	Description
width	int	-	-	Camera output frame width
height	int	-	-	Camera output frame height
fps	int	-	-	Camera output fps
max_zoom_factor	decimal	-	-	Maximum digital zoom factor
model_path	string	-	-	Object detection model in RKNN format, created with rknn-toolkit2
i2c_device	string	-	-	I2C peripheral where PCA9685 module is connected
i2c_addr	int	-	-	I2C address of PCA9685 module
pan_max	int	0	-	Pan angle max value
pan_min	int	0	-	Pan angle min value
tilt_max	int	0	-	Tilt angle max value
tilt_min	int	0	-	Tilt angle min value
pan_base	int	0	-	Pan angle default value
tilt_base	int	0	-	Tilt angle default value
pan_kp	float	-	-	Pan PID controller kp coefficient
pan_ki	float	-	-	Pan PID controller ki coefficient
pan_kd	float	-	-	Pan PID controller kd coefficient
tilt_kp	float	-	-	Tilt PID controller kp coefficient
tilt_ki	float	-	-	Tilt PID controller ki coefficient
tilt_kd	float	-	-	Tilt PID controller kd coefficient
gimbal_deadzone	float	0.0	1.0	Target tracking deadzone threshold applied to normalized error in range [-1:1]
auto_return_to	int	0	-	Time before returning to default pan and tilt angle when tracked target is lost

Table 1: Configuration parameters

Field	Type	Min	Max	Description
osd_enabled	bool	-	-	On screen display status at startup
osd_color	list	-	-	OSD text color in BGR format as list
osd_thickness	int	-	-	OSD text thickness
osd_scale	int	-	-	OSD text scale
osd_pos_x	int	-	-	OSD X start position
osd_pos_y	int	-	-	OSD Y start position

Table 2: Configuration parameters, continue

3 Communication protocol

The Smartvision System and the Smartvision Client exchange data with a set of JSON messages and UDP commands.

3.1 JSON Messages

JSON messages are sent by the server application to notify the presence on the network end updates the current state. There are two messages, discovery message and status message.

3.1.1 Discovery Message

Discovery message is sent by the server every 5 s and has the following structure:

```
{  
  "device": "multivision_system",  
  "ip": "0.0.0.0",  
  "rtsp": "8554",  
  "cmd": 5005  
}
```

Where:

- **device**: string, is the application name, equal to "multivision_system";
- **ip**: string, is the server ip address used for RTSP vide and UDP commands;
- **rtsp**: string, is the RTSP video port;
- **cmd**: int, is the UDP server PORT.

So, the RTSP video is available at address:

ip : rtsp / video

And UDP commands must be sent to:

ip : cmd

3.1.2 Status Message

The status message is sent by the server every 500 ms and has the following structure:

```
{
  "zoom":0.0,
  "pan":90,
  "tilt":90,
  "autoZoom":true,
  "osdEnabled":true,
  "recording":false
}
```

Where:

- **zoom**: decimal, current zoom factor;
- **pan**: int, current pan angle;
- **tilt**: int, current tilt angle;
- **autoZoom**: boolean, auto zoom function during target tracking enabled or disabled;
- **osdEnabled**: boolean, on screen info enabled or disabled;
- **recording**: video recording function enabled or disabled.

3.2 Command list

A set of UDP commands can be used to control the Smartvision System. Each packet has a fixed length of 5 bytes where the first byte is the command ID.

3.2.1 0x01 - SET ZOOM

Field	Name	Description
0	cmd_id	Command id, equal to 0x01
1	zoom_integer	Desired zoom integer part
2	zoom_decimal	Desired zoom decimal part
3	-	Empty
4	-	Empty

Table 3: Set zoom command

3.2.2 0x02 - SET GIMBAL

Field	Name	Description
0	cmd_id	Command id, equal to 0x02
1	move_pan	43: Increase pan angle by 5 45: Decrease pan angle by 5
2	move_tilt	43: Increase tilt angle by 5 45: Decrease tilt angle by 5
3	reset	1: Reset pan and tilt angles
4	-	Empty

Table 4: Set gimbal command

3.2.3 0x03 - CAMERA CONTROL

Field	Name	Description
0	cmd_id	Command id, equal to 0x03
1	camera_operation	1: Take snapshot 2: Toggle video recording
2	enable_osd	1: Enable OSD 0: Disable OSD
3	-	Empty
4	-	Empty

Table 5: Camera control command

3.2.4 0x04 - SET IMAGE TYPE

Field	Name	Description
0	cmd_id	Command id, equal to 0x04
1	image_type	0: Visible 1: Thermal 2: Mixed
2	-	Empty
4	-	Empty
4	-	Empty

Table 6: Set image type command

3.2.5 0x05 - TARGET TRACKING

Field	Name	Description
0	cmd_id	Command id, equal to 0x05
1	target_x	Target X coordinate normalized in range [0:255]
2	target_y	Target Y coordinate normalized in range [0:255]
3	-	Empty
4	-	Empty

Table 7: Target tracking command

3.2.6 0x06 - TRACKING MODE

Field	Name	Description
0	cmd_id	Command id, equal to 0x06
1	auto_zoom	0: Disable auto zoom during target tracking 1: Enable autozoom during target tracking
2	-	Empty
3	-	Empty
4	-	Empty

Table 8: Tracking mode command